

Technical Documentation: Village Locator Application

1. Project Overview

The Village Locator is a full-stack web application designed to help users search for and locate villages, sub-districts, and districts across various states. The application processes very large demographic datasets (e.g., CSV files exceeding 1.6 GB) and provides a fast, responsive user interface to search through this data and display geographic locations using Google Maps integration.

2. Architecture & Technology Stack

The project adopts a modern monolithic repository structure with decoupled frontend and backend logic.

2.1 Backend / API Layer

- **Framework:** Next.js (App Directory)
- **Language:** TypeScript
- **Database:** PostgreSQL
- **ORM:** Prisma (prisma-client-js)
- **Key Modules:**
 - csv-parser for streaming and seeding large demographic CSV files.
 - Next.js API Routes (app/api) to handle RESTful client requests.

2.2 Frontend / Client Layer

- **Framework:** React 19 via Vite
- **Language:** TypeScript
- **Styling:** Tailwind CSS (configured at the root level via PostCSS)
- **Features:**
 - Dynamic Search Logic
 - Google Maps Modal for village directions.

3. Database Schema (Prisma)

The database structure is normalized to reflect the geographic hierarchy:

1. **State:**
 - Fields: id, code (Unique), name
 - Relations: One-to-many with District
2. **District:**
 - Fields: id, code (Unique), name, stateId
 - Relations: One-to-many with SubDistrict
3. **SubDistrict:**
 - Fields: id, code (Unique), name, districtId
 - Relations: One-to-many with Village
4. **Village:**
 - Fields: id, code (Unique), name, fullAddress, subDistrictId

4. Key Design Decisions

- **Separation of Concerns:** The client interface is hosted under the /client directory (Vite), while server-side rendering and API endpoints leverage the Next.js framework in the root.
- **Large Data Processing:** The backend seeding script (prisma/seed.ts) uses streams to parse multi-gigabyte CSV datasets without exceeding memory limits.
- **Ignored Artifacts:** .env, .next, node_modules, and heavy raw CSV datasets are explicitly excluded from production builds and Git history to maintain security and portability.